# CONDITIONAL PROBABILITY & INDEPENDENCE

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February 2, 2024

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# What Is Conditional Probability?

- **Conditional Probability:** The probability of an event when some partial information about the experiment is available.
  - Consider the probability that it will rain tomorrow.
  - Now, consider the probability that it will rain tomorrow **given that it has rained today?**
  - Observing rain today will change the anticipation on tomorrow's weather.
- Even when no partial information is available, conditional probability is a concept that can be used to compute the desired probabilities more easily.

# Understanding Conditional Probability

- Suppose we toss 2 fair dice.
  - Each of the 36 possible outcomes is equally likely to occur and hence has probability $\frac{1}{36}$.
- What is the probability that the sum of the upturned faces equals 8?
  - The answer is $\frac{|E|}{|S|} = \frac{5}{36}$ since:

$$E = \{(2,6),(3,5),(4,4),(5,3),(6,2)\}$$

# Understanding Conditional Probability (cont'd)

- Now, suppose, we toss 2 fair dice.
  - An observer tells us that the the first die shows a 3.
- Now, what is the probability that the sum of the upturned faces equals 8?
  - With the partial information, the sample space reduces to:

$$S = \{(3,1),(3,2),(3,3),(3,4),(3,5),(3,6)\}$$

  - The desired event is, now, only possible with a single outcome.

$$E = \{(3,5)\}$$

  - Thus, conditional on the partial information, the probability of this event becomes $\frac{|E|}{|S|} = \frac{1}{6}$.

# Calculating Conditional Probabilities

- If we observe that event $F$ has already occurred, the probability of event $E$ is denoted with:

$$P(E|F)$$

an read as "probability of $E$ given $F$".

- The conditional probability is calculated with the following formula:

$$P(E|F) = \frac{P(E \cap F)}{P(F)}$$

- Note that, $P(F) \neq 0$ must be satisfied. Otherwise, it would not have been observed.

# Calculating Conditional Probabilities: Example

Q1. We toss two dice. Suppose that somebody has told the sum is 8. What is the probability that the upturned faces show the same value?

# Calculating Conditional Probabilities: Example (cont'd)

A1. We have the following information:

- The experiment originally has 36 equally likely outcomes.
- $E$: The event that the upturned faces show the same value.

$$E = \{(1,1),(2,2),(3,3),(4,4),(5,5),(6,6)\}$$

- $F$: The event that the sum is 8.

$$F = \{(2,6),(3,5),(4,4),(5,3),(6,2)\}$$

The question asks $P(E|F) = \frac{P(E\cap F)}{P(F)}$.

$$P(F) = \frac{5}{36}, \quad P(E\cap F) = P(\{(4,4)\}) = \frac{1}{36}$$

$$P(E|F) = \frac{P(E\cap F)}{P(F)} = \frac{\frac{1}{36}}{\frac{5}{36}} = \frac{1}{5}$$

# Calculating Conditional Probabilities: Example II

Q2. In a university, 62 percent of the students own a iPhone, while 44 percent of these students own a LG mobile phone. It is known that 20 percent of the students own neither of these mobile phones. We randomly select a student.

a. Given that the student owns a LG mobile phone, what is the probability that the student also owns a iPhone.
b. Given that the student does not own an iPhone, what is the probability that the student owns a LG Mobile phone.
c. Given that the student owns at least one of these mobile phones, what is the probability that the student owns both mobile phones.

# Calculating Conditional Probabilities: Example II (cont'd)

A2. Let us define the following events:

- $I$: The randomly selected student owns an iPhone,
- $L$: The randomly selected student owns an LG mobile phone.

The question gives us the following information:

$$P(I) = 0.62, \quad P(L) = 0.44, \quad P((I \cup L)^c) = 0.20$$

We can also find the following values:

- The prob. that the student owns at least one of these phones:

$$P(I \cup L) = 1 - P((I \cup L)^c) = 0.80$$

- The prob. that the student owns both of these phones:

$$P(I \cap L) = P(I) + P(L) - P(I \cup L) = 0.62 + 0.44 - 0.80 = 0.26$$

# Calculating Conditional Probabilities: Example II (cont'd)

a.

$$P(I|L) = \frac{P(I \cap L)}{P(L)} = \frac{0.26}{0.44} = \frac{13}{22}$$

b.

$$P(L|I^c) = \frac{P(L \cap I^c)}{P(I^c)} = \frac{P(L) - P(L \cap I)}{1 - P(I)} = \frac{0.44 - 0.26}{0.38} = \frac{0.18}{0.38} = \frac{9}{19}$$

c.

$$P(I \cap L|I \cup L) = \frac{P((I \cap L) \cap (I \cup L))}{P(I \cup L)} = \frac{P(L \cap I)}{P(I \cup L)} = \frac{0.26}{0.80} = \frac{13}{40}$$

# Product Rule

- Using the definition of conditional probability, we can also calculate the probability of intersection of events:

$$P(E|F) = \frac{P(E \cap F)}{P(F)} \Rightarrow P(E \cap F) = P(F)P(E|F) = P(E)P(F|E)$$

- In other words, the probability of two events occuring at the same time is calculated by multiplying:
  - the probability of the first event
  - and the probability of the second event, given that the first event occurs.

# Product Rule: Example

Q3. Suppose that an urn contains 8 red balls and 4 blue balls and we draw two balls from the urn without replacement. If we assume that, at each draw, each ball in the urn is equally likely to be chosen:

a. What is the probability that both balls drawn are red?
b. What is the probability that both balls drawn are blue?
c. What is the probability that the balls are of different colors?

A3. Although combination terminology could have been used to solve this question, we will use conditioning.

Let $R_1$ and $R_2$, respectively, denote the events that the first and the second balls drawn are red. We have the following information:

$$P(R_1) = \frac{8}{12} = \frac{2}{3}, \quad P(R_1^c) = \frac{4}{12} = \frac{1}{3}$$

# Product Rule: Example (cont'd)

a. The question asks for $P(R_1 \cap R_2)$.
   - We know that $P(R_1) = \frac{2}{3}$.
   - Given that the first ball is red, the probability of the second ball drawn being red is:
     $$P(R_2|R_1) = \frac{7}{11}$$
     since, after the first ball drawn is observed to be red, there will be 11 balls left in the urn, and 7 of them will be red.
   - The desired probability is:
     $$P(R_1 \cap R_2) = P(R_1)P(R_2|R_1) = \frac{2}{3}\frac{7}{11} = \frac{14}{33}$$

# Product Rule: Example II (cont'd)

b. The question asks for $P\left(R_1^c \cap R_2^c\right)$.

- We know that $P\left(R_1^c\right) = \frac{1}{3}$.
- Given that the first ball is blue, the probability of the second ball drawn being blue is:

$$P\left(R_2^c | R_1^c\right) = \frac{3}{11}$$

since, after the first ball drawn is observed to be blue, there will be 11 balls left in the urn, and 3 of them will be blue.

- The desired probability is:

$$P\left(R_1^c \cap R_2^c\right) = P\left(R_1^c\right) P\left(R_2^c | R_1^c\right) = \frac{1}{3}\frac{3}{11} = \frac{1}{11}$$

# Product Rule: Example II (cont'd)

c. The question asks for $P(R_1^c \cap R_2) + P(R_1 \cap R_2^c)$.

$$P(R_1 \cap R_2^c) = P(R_1)P(R_2^c|R_1) = \frac{8}{12}\frac{4}{11} = \frac{8}{33}$$
$$P(R_1^c \cap R_2) = P(R_1^c)P(R_2|R_1^c) = \frac{4}{12}\frac{8}{11} = \frac{8}{33}$$

Thus:

$$P(R_1 \cap R_2^c) + P(R_1^c \cap R_2) = \frac{8}{33} + \frac{8}{33} = \frac{16}{33}$$

# Total Probability Rule

- We can use the conditional probabilities to calculate the probability of a certain event by conditioning it on another.
- Suppose we want to calculate $P(E)$, and we have the information on "the probability" of another event $F$.
  - The event $E$ can be expanded as:
$$E = (E \cap F) \cup (E \cap F^c)$$
  - $(E \cap F)$ and $(E \cap F^c)$ are mutually exclusive events (disjoint) events.
- Therefore:
$$\begin{aligned} P(E) &= P(E \cap F) + P(E \cap F^c) \\ &= P(F)P(E|F) + P(F^c)P(E|F^c) \end{aligned}$$

# Total Probability Rule: Example

Q4. An insurance company believes that people can be divided into two classes: those who are accident prone and those who are not. The company's statistics show that an accident-prone person will have an accident at some time within a fixed 1-year period with probability $0.4$, whereas this probability decreases to $0.2$ for a person who is not accident prone. If we assume that $30\%$ of the population is accident prone, what is the probability that a new policyholder will have an accident within a year of purchasing a policy?

# Total Probability Rule: Example (cont'd)

A4. We have the following information:

- $A$: The new policy holder is accident prone, $P(A) = 0.3$.
- $A^c$: The new policy holder is **not** accident prone,
$$P(A^c) = 1 - P(A) = 0.7$$
- $B$: The new policy holder will have an accident within a year.
$$P(B|A) = 0.4, \quad P(B|A^c) = 0.2$$

The question asks $P(B)$. We condition this probability on the event of being accident-prone:

$$\begin{aligned} P(B) &= P(A)P(B|A) + P(A^c)P(B|A^c) \\ &= (0.3)(0.4) + (0.7)(0.2) = 0.26 \end{aligned}$$

# Total Probability Rule: Example II

Q5. Bowl A contains three red and two blue chips, and bowl B contains four red and three blue chips. A chip is drawn at random from bowl A and transferred to bowl B. Compute the probability of then-drawing a red chip from bowl B.

# Total Probability Rule: Example II

Q5. Bowl A contains three red and two blue chips, and bowl B contains four red and three blue chips. A chip is drawn at random from bowl A and transferred to bowl B. Compute the probability of then-drawing a red chip from bowl B.

A5. The desired probability depends on what you have transferred to bowl B in the first place. Let $R_1$ denote the event that a red chip is transferred from bowl A to bowl B. Note that:

$$P(R_1) = 0.6, \quad P(R_1^c) = 0.4$$

# Total Probability Rule: Example II (cont'd)

A5. Let $R_2$ denote the event that then-drawing a red chip from bowl B. We have the following conditional probabilities

$$P(R_2|R_1) = \frac{5}{8} = 0.625, \quad P(R_2|R_1^c) = \frac{4}{8} = 0.5$$

because, if we transfer a red chip in the first draw, there will be 5 out of 8 chips in the second bowl, otherwise, there will be 4 red chips out of 8 chips.

# Total Probability Rule: Example II (cont'd)

A5. Let $R_2$ denote the event that then-drawing a red chip from bowl B. We have the following conditional probabilities

$$P(R_2|R_1) = \frac{5}{8} = 0.625, \quad P(R_2|R_1^c) = \frac{4}{8} = 0.5$$

because, if we transfer a red chip in the first draw, there will be 5 out of 8 chips in the second bowl, otherwise, there will be 4 red chips out of 8 chips.

We can conclude:

$$\begin{aligned} P(R_2) &= P(R_1)P(R_2|R_1) + P(R_1^c)P(R_2|R_1^c) \\ &= 0.6 \times 0.625 + 0.4 \times 0.5 = 0.575 \end{aligned}$$

# What Is Independence of Events?

- In some cases, the probability of event $E$ does not change on the fact that the other event $F$ occurs or not.

$$P(E) = P(E|F) = P(E|F^c)$$

- For example, the probability of rainy weather does not change with respect to:
  - the outcome of a rolled die,
  - or the outcome of a coin flip.
- If $P(E) = P(E|F)$, then the probability of two events occurring at the same time: $P(E \cap F) = P(F)P(E|F) = P(E)P(F)$
- **Independent Events:** Two events are said to be independent if and only if:

$$P(E \cap F) = P(E)P(F)$$

# Understanding Independence of Events

- Independence of events is not something that you can decide. If you want to prove that two events (say $E$ and $F$) are independent, you need to show that:

$$P(E \cap F) = P(E)P(F)$$

- Independence is a symmetric relationship:
  - When $E$ is independent of $F$, we say $F$ is independent of $E$ too.
  - It does not matter which event is supposed to occur first over the time line.
- If $E$ is independent of $F$, $E$ is also independent of $F^c$.

# Independent Events: Example

Q6. Suppose we toss two dice, sequentially. $E$ is the event that the first die comes $4$. $F$ is the event that the sum of two dice is 7. Are these events independent?

# Independent Events: Example (cont'd)

A6. Let us check the independence condition:

- $P(E) = P(\{(4,1),(4,2),(4,3),(4,4),(4,5),(4,6)\}) = \frac{6}{36}$
- $P(F) = P(\{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}) = \frac{6}{36}$
- $P(E \cap F) = P(\{(4,3)\}) = \frac{1}{36}$.

Since $P(E \cap F) = P(E)P(F)$, these two events are **independent.**

# Independent Events: Example II

Q7. Suppose we toss two dice, sequentially. $E$ is the event that the first die comes $4$. $F$ is the event that the sum of two dice is 6. Are these events independent?

# Independent Events: Example II (cont'd)

A7. Let us check the independence condition:

- $P(E) = P(\{(4,1),(4,2),(4,3),(4,4),(4,5),(4,6)\}) = \frac{6}{36}$
- $P(F) = P(\{(1,5),(2,4),(3,3),(4,2),(5,1)\}) = \frac{5}{36}$
- $P(E \cap F) = P(\{(4,2)\}) = \frac{1}{36}$.

Since $P(E \cap F) \neq P(E)P(F)$, these two events are **not independent.**

# Independent Events: Example III

Q8. In a community, a family:

- owns a TV with probability $0.8$
- owns a car with probability $0.6$

If owning a TV and owning a car are independent events, what is the probability that family owns none of these things?

# Independent Events: Example III (cont'd)

A8. The given information is as follows:

- $T$: Event of owning a TV, $P(T) = 0.8$.
- $C$: Event of owning a car, $P(C) = 0.6$.

$T \cap C$: Event of owning both. Due to independence:

$$P(T \cap C) = P(T)P(C) = (0.8)(0.6) = 0.48$$

$T \cup C$: Event of owning at least one of these things:

$$P(T \cup C) = P(T) + P(C) - P(T \cap C) = 0.8 + 0.6 - 0.48 = 0.92$$

$(T \cup C)^c$: Event of owning none of these things:

$$P((T \cup C)^c) = 1 - P(T \cup C) = 1 - 0.92 = 0.08$$

# Bayes' Formula

- **Bayes' Theorem:** Let us expand the conditional probability formula:

$$P(E|F) = \frac{P(E \cap F)}{P(F)} = \frac{P(E)P(F|E)}{P(E)P(F|E) + P(E^c)P(F|E^c)}$$

- The right hand side of the formula is called the **Bayes' Formula**[1] and useful in calculating the probability of prior events given posterior information.
- The best way to understand **Bayes' Formula** is to solve related questions.

[1] Thomas Bayes, 1701-1761.

# Bayes' Formula: Example

Q9. Consider two urns:

- The first contains two red and seven blue balls,
- The second contains five red and six blue balls.

A student flips a fair coin and draws a ball from the first or the second urn depending on whether the outcome was heads or tails. What is the conditional probability that the outcome of the toss was heads given that a red ball was selected?

# Bayes' Formula: Example (cont'd)

A9. The given information is as follows:

- $H$: The event that coin flip shows heads, $P(H) = 0.5$.
- $H^c$: The event that coin flip shows tails.

$$P(H^c) = 1 - P(H) = 0.5$$

- $R$: The event that a red ball is selected. Notice that:

$$P(R|H) = \frac{2}{9}, \quad P(R|H^c) = \frac{5}{11}$$

The question asks for $P(H|R)$. According to Bayes' Formula:

$$\begin{aligned} P(H|R) &= \frac{P(H)P(R|H)}{P(H)P(R|H)+P(H^c)P(R|H^c)} \\ &= \frac{\left(\frac{1}{2}\right)\left(\frac{2}{9}\right)}{\left(\frac{1}{2}\right)\left(\frac{2}{9}\right)+\left(\frac{1}{2}\right)\left(\frac{5}{11}\right)} = \frac{22}{67} \end{aligned}$$

# Bayes' Generalized Formula

- In the previous form of Bayes' formula, the sample space was partitioned in two disjoint sets $E$ and $E^c$.

$$P(E|F) = \frac{P(E)P(F|E)}{P(E)P(F|E) + P(E^c)P(F|E^c)}$$

- In general, the event space can be partitioned in any countable number of disjoint sets $E_1, E_2, \ldots$. Then, it is useful to expand the denominator as:

$$\sum_j P(E_j)P(F|E_j)$$

and generalize the Bayes' formula as:

$$P(E_i|F) = \frac{P(E_i)P(F|E_i)}{\sum_j P(E_j)P(F|E_j)}$$

# Bayes' Generalized Formula: Example

Q10. There are four coins in a box. One is a fair coin, another is a two-headed coin, and the last two are biased coins that come up heads with probability 0.75. We select one of the four coins randomly and flip it without looking at it; it shows heads. What is the probability that it was the fair coin?

A10. Let us define the following events:

- $H$: Observing a head after the coin flip.
- $C_1$: Selecting the fair coin.

$$P(C_1) = 0.25, \quad P(H|C_1) = 0.5$$

- $C_2$: Selecting the two-headed coin, $P(C_2) = 0.25$.

$$P(C_2) = 0.25, \quad P(H|C_2) = 1$$

- $C_3$: Selecting a head-biased coin, $P(C_3) = 0.5$.

$$P(C_3) = 0.5, \quad P(H|C_3) = 0.75$$

# Bayes' Generalized Formula: Example (cont'd)

A10. The question asks for $P(C_1|H)$. By using Bayes' generalized formula:

$$\begin{aligned} P(C_1|H) &= \frac{P(C_1 \cap H)}{P(H)} \\ &= \frac{P(C_1)P(H|C_1)}{P(C_1)P(H|C_1)+P(C_2)P(H|C_2)+P(C_3)P(H|C_3)} \\ &= \frac{(0.25)(0.5)}{(0.25)(0.5)+(0.25)(1)+(0.5)(0.75)} \\ &= \frac{0.125}{0.75} = \frac{1}{6} \end{aligned}$$

END OF LECTURE SLIDE 03